

Parth Mital

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🐙 github.com/parthmital 🌐 parthmital-portfolio.vercel.app/

Education

Vellore Institute of Technology (VIT)

B.Tech in Computer Science & Engineering (CGPA: 8.75 / 10)

2023 – 2027

Vellore, India

Projects

🎧 Muse: Self-Hosted AI Music Streaming Platform

- Built **full-stack music streaming platform** with **redesigned UI**, playlists, search, radio mode & persistent playback.
- Developed **hybrid recommender** using embeddings + audio features with **diversity-aware ranking**.
- Enabled **lossless audio streaming** with adaptive bitrate/quality selection, continuous playback & smooth transitions.
- Optimised **recommendation latency** using caching, background jobs & precomputed user profiles.
- **Tech Stack:** Node.js, TypeScript, Fastify, Python, FastAPI, PyTorch, SentenceTransformers, FAISS, SQLite (WAL), Next.js, React, TailwindCSS, SWR, Dash.js, Zod, Pino, Tidal API, Last.fm API, Spotify Web API

🎧 VIT Resource Automation Pipeline - Academic Resource Lifecycle Toolkit

- Built an **end-to-end automation pipeline** processing **6,000+ academic documents** per semester, reducing collection time from **weeks** to under **1 hour**, including asynchronous Selenium/Playwright scrapers for **PYQs**.
- Automated **bulk data extraction** from VTOP/UniBud across **150+ faculty** & **6 concurrent subjects** and implemented an **analysis engine** (pypdf + COM) to rank faculty by slide count & content depth.
- **Standardized** legacy files (.doc, .ppt) into 100% PDF format for consistent access.
- **Tech:** Python, Selenium, Playwright, pypdf, pywin32, comtypes, Chrome DevTools Protocol, CSV.

🎧 VIT Carpool - Full-Stack Campus Ride-Sharing Platform | 🌐 vitcarpool.vercel.app

- Built **centralized ride-sharing platform** replacing fragmented WhatsApp-based coordination.
- Implemented **structured filtering system** (source, destination, time) for **real-time** efficient ride discovery.
- Achieved **2,600+ views/impressions** within the VIT student community, indicating strong engagement.
- Secured access via domain-restricted **Google OAuth** & enforced data isolation using **Supabase RLS**.
- **Tech:** React 18, TypeScript, Vite, Tailwind CSS, Supabase, PostgreSQL, REST APIs, Google OAuth.

Technical Skills

Languages: C, C++, Java, Python, TypeScript, JavaScript

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Digital Systems Design, Microprocessors & Microcontrollers

Software Development: Full-Stack Development, Frontend, Backend, Databases, REST APIs, Authentication

AI/ML & Data Science: Machine Learning, Deep Learning, Model Training & Evaluation, Computer Vision, NLP, Feature Engineering, EDA, Data Processing, ETL Pipelines

Cloud & DevOps: Containerization, Cloud Deployment, AWS, Docker

Experience

Hiraeth Tech

Web Development Intern

Jun 2025 – Jul 2025

Vadodara, India

- Built **5+ responsive WordPress websites** using Elementor & custom CSS.
- Optimised images & assets to improve page load performance.
- Ensured cross-device responsiveness & consistent UI/UX.